

B.Sc. (Data Science)

DS101T : Problem Solving and Python Programming

Unit 2: Python Libraries

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Pandas

- Pandas is a popular **open-source library** used for **data manipulation and analysis in Python**.
- It provides **easy-to-use** functions needed to work with **structured data** seamlessly.
- Pandas also **integrates** seamlessly with other popular Python libraries, such as **NumPy for numerical computing** and **Matplotlib for data visualization**. This makes it a powerful asset for **data driven tasks**.
- Pandas **excels** in handling missing data, reshaping datasets, merging and joining multiple datasets, and performing complex operations on data, making it exceptionally useful for **data cleaning and manipulation**.
- It is built on top of NumPy.
- pandas stands for *tabular heterogeneous* data. Where , NumPy is best suited for working with homogeneous numerical data.

How to Import the Necessary Libraries

- To save time and typing, we often

`import Pandas as pd`

shorter `pd.read_csv()` instead of `pandas.read_csv()` for reading CSV files, making our code **more efficient and readable**.

Pandas Data Structures

- Pandas introduces two key data structures:

Series

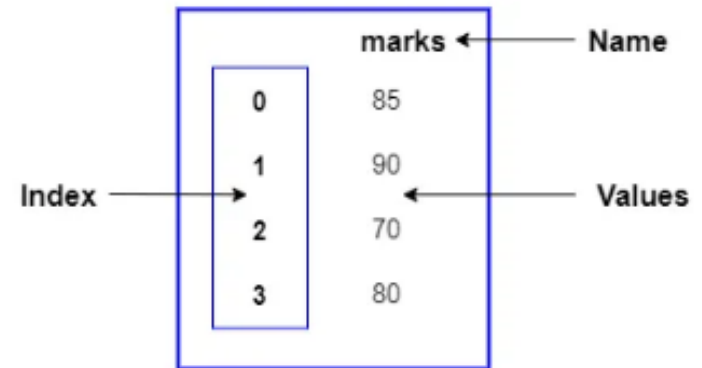
DataFrame

- These structures allow users to manipulate, clean, and analyze datasets efficiently.
- **A Series** is a **one-dimensional array**-like object that can **hold any data type**.
- **A DataFrame** is a **two-dimensional table** with labeled axes (rows and columns).
- **Series** — 1D
- **DataFrame** — 2D
- **Panel** — 3D

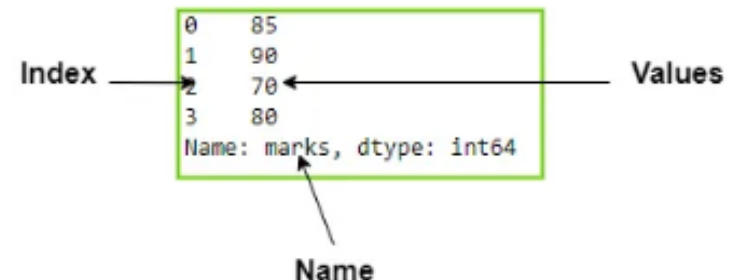
Data Structure	Dimensionality	Format	View
Series	1D	Column	name 0 Rukshan 1 Prasadi 2 Gihan 3 Hansana age 0 25 1 25 2 26 3 24 marks 0 85 1 90 2 70 3 80
DataFrame	2D	Single Sheet	name age marks 0 Rukshan 25 85 1 Prasadi 25 90 2 Gihan 26 70 3 Hansana 24 80
Panel	3D	Multiple Sheets	name age marks 0 Rukshan 25 85 1 Prasadi 25 90 2 Gihan 26 70 3 Hansana 24 80

The pandas Series

- The ***Series*** object represents **one-dimensional data** structures in Pandas.
- **Key points:**
 - 1-d array with Homogeneous data
 - Size Immutable
 - Values of Data Mutable
- A series consists of two components.
 - One-dimensional data (Values)
 - Index



`pd.Series([85, 90, 70, 80], name='marks')`



Series creation

- **Syntax:**
- `pd.Series(data=..., index=..., dtype=..., name=...)`
- To create a series, use **Series()** class constructor and pass as an argument containing the data to be included in it. Here, ***data*** can be one of the following:
 - A one-dimensional ndarray
 - A Python list
 - A Python dictionary
 - A scalar value
- If an index is not specified, the default index **[0,... n-1]** will be created, where **n** is the length of the data. A series can be created in a variety of ways.

```
import pandas as pd
data = ['Steve', '35', 'Male', '3.5']
series = pd.Series(data, index=['Name', 'Age', 'Gender', 'Rating'])
print(series)
```

Output:

Name Steve

Age 35

Gender Male

Rating 3.5

dtype: object

DataFrame

- A DataFrame is a two-dimensional size-mutable, potentially heterogeneous tabular data structure composed of rows and columns.
- **Key Points:**
 - Each column of a DataFrame is a pandas Series.
 - These columns should be of the same length, but they can be of different data types — float, int, bool, and so on.
 - DataFrames are both value and size-mutable .
- A DataFrame is a two-dimensional labeled data structure with columns that can hold different data types. It is similar to a table in a database or a spreadsheet.

Columns

Rows

Data

	<i>Name</i>	<i>Team</i>	<i>Number</i>	<i>Position</i>	<i>Age</i>
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Creating a Pandas DataFrame

- Pandas DataFrame will be created by loading the datasets from existing storage, storage can be SQL Database, CSV file, and Excel file. Pandas DataFrame can be created from the lists, dictionary, and from a list of dictionary etc.

```
# Take a 2D array as input to your DataFrame my_2darray = np.array([[1, 2, 3], [4, 5, 6]])
print(pd.DataFrame(my_2darray)) # Take a dictionary as input to your DataFrame
my_dict = {1: ['1', '3'], 2: ['1', '2'], 3: ['2', '4']} print(pd.DataFrame(my_dict)) # Take a
DataFrame as input to your DataFrame my_df = pd.DataFrame(data=[4,5,6,7],
index=range(0,4), columns=['A']) print(pd.DataFrame(my_df)) # Take a Series as input to
your DataFrame my_series = pd.Series({"Belgium":"Brussels", "India":"New Delhi",
"United Kingdom":"London", "United States":"Washington"}) print(pd.
DataFrame(my_series))
```

Take a 2D array as input to your DataFrame

```
my_2darray = np.array([[1, 2, 3], [4, 5, 6]])  
print(pd.DataFrame(my_2darray))
```

This example creates a DataFrame from a 2D NumPy array `my_2darray` using the `pd.DataFrame()` function. The array is passed as an argument to the function and the resulting DataFrame is printed using the `print()` function.

Output:

```
   0  1  2  
0   1  2  3  
1   4  5  6
```

Take a dictionary as input to your DataFrame

```
my_dict = {1: ['1', '3'], 2: ['1', '2'], 3: ['2', '4']}  
print(pd.DataFrame(my_dict))
```

This example creates a DataFrame from a dictionary `my_dict`. The keys of the dictionary become the column names and the values become the data in the columns. The resulting DataFrame is printed using the `print()` function.

Output:

```
   1  2  3  
0   1  1  2  
1   3  2  4
```

Take a DataFrame as input to your DataFrame

```
my_df = pd.DataFrame(data=[4,5,6,7], index=range(0,4), columns=['A']) print(pd.  
DataFrame(my_df))
```

This example creates a DataFrame from an existing DataFrame my_df. The data parameter is set to the values of the original DataFrame, the index parameter is set to the range of indices of the original DataFrame, and the columns parameter is set to the column names of the original DataFrame. The resulting DataFrame is printed using the print() function.

Output:

	A
0	4
1	5
2	6
3	7

Take a Series as input to your DataFrame

```
my_series = pd.Series({"Belgium":"Brussels", "India":"New Delhi", "United Kingdom" :  
"London", "United States":"Washington"})  
print(pd.DataFrame(my_series))
```

This example creates a DataFrame from a Pandas Series my_series. The keys of the Series become the index of the DataFrame and the values become the data in the DataFrame. The resulting DataFrame is printed using the print() function.